Release Notes

BASE24 Interac Interfaces
with
Triple DES Enhancements

BASE24-imni/imnsc
Release 6.0v10

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ACI Worldwide (Canada) Inc.
200 Wellington St. W. Suite 700
Toronto, Ontario, M5V 3C7
(416) 813-3000

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Toronto, Ontario, M5V 3C7
(416) 813-3000

REVISION HISTORY

Record of Changes

Author	Date	Version	Comments
ACI	November 2011	1.0	Initial version

Reference Documents

Document Name	Author	Version/Date
ACI KEK Change Procedures using TSS	ACI	Version 2.2, March 2010
Addendum of ACI KEK Change Procedures using TSS for Adoption of Triple DES Cryptography	ACI	November 2011
Interac SIG Atalla calls used for Triple DES BASE24-imni/imnsc Interac Interfaces Release 6.0	ACI	Version 1.0, October 2011

Audience

This document is intended for the BASE24 system professionals who are adequately trained and/or experienced for the contents covered in this document, due to requirements for specific knowledge and understanding of acronyms.

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SECTION 1 INTRODUCTION

This release of the BASE24 Interac Interface software package includes Interac Triple DES (TDES) enhancements, associated documents and peripheral elements.

The Interac TDES adoption will be phased as follows:

1. Phase 1 – Apply TDES cryptography to the Key Exchange Keys (KEK).
   Members apply TDES values to their Key Exchange Keys (KEK) in this phase.

2. Phase 2 – Apply TDES cryptography to all working keys.
   After phase 1 is successfully implemented, members apply TDES values to the working keys (PIN, MAC and MSG where applicable).

The benefit of implementing TDES KEKs prior to the TDES working keys is that it enables the Members to complete the manual KEK exchange process first and confirm the TDES KEKs are implemented properly. This helps minimize the risk of the TDES working key migration.

Delivery Items

Customers will continue to use standard procedures to acquire this Interac TDES package from your customer manager.

A) Standard BASE modules under BASE24 R6.V10

B) Standard BASE24 TSS R9.2, Service Pack 6

C) Canadian BASE24-imni/imnsc R6.V10

D) Associated documentation, see the list below:

Reference Number	Items	Delivered	Memo
D1	Release Notes		This document
D2	BASE24 IMNI/IMNSC Installation Guide V7		Installation Guide
D3	Addendum of ACI KEK Change Procedures for TDES V2.2		TSS
D4	Interac SIG Atalla calls used for TDES V1.0		Atalla Calls

Standard ACI product manuals are available on the ACI Infolink site. The Interac TDES specific documents are included in this delivery package.

TSS TDES Reference: BASE24 Transaction Security Services Processing Guide Release 6.0, Version 10.

Prerequisites

- The new BASE24 Interac interface package for TDES has been acquired;

- Phase 1 - Transaction security Key Exchange Key(s) will be upgraded to double length which allows for TDES encryption.

    1) An adequate backup of the existing system has been exercised;

    2) Atalla Hardware Security Module (HSM)
        a) Ensure its release level in use should be fully supporting TDES cryptography for KEK. Please contact your Atalla vendor for more detail;
        b) Ensure all the required banking commands/options related to activities with double length Key Exchange Keys have been enabled; For more details, please refer to reference document <D4>.

    3) An agreement has been obtained from each counterpart member to enter this phase;

    4) A manual exchange of double length Key Exchange Keys with each counterpart member has been completed.

- Phase 2 - Transaction security working key(s) will be upgraded to double length.

    1) Phase 1 is proven successful;

    2) An agreement has been obtained from each counterpart member to enter this phase.

    3) Atalla Hardware Security Module (HSM)
        a) Ensure the Atalla HSM release level in use fully supports TDES cryptography for KEK and working keys. Please contact your Atalla vendor for more detail;
        b) Ensure all the required banking commands/options related to activities with double length Key Exchange Keys and double length working keys have been enabled; For more details, please refer to reference document <D4>.
        c) AKB mode is employed;
        Note: For more details on AKB conversion, refer to Section 4 of the BASE24 TSS Processing Guide.

SECTION 2: IMPLEMENTATION GUIDELINES

Implementation Steps

General

Note: ACI recommends this step is performed prior to Phase 1. However, the customer may choose to install the new TDES package before Phase 2.

Install the new BASE24 Interac interface package for TDES:

1) Build the BASE24-imni/imnsc system with TDES enhancement under BASE24 R6.V10.
    a. There are two options to introduce BASE24 R6.V10:
    (1) A complete upgrade of BASE24 to R6.V10
    or
    (2) A complete installation of the BASE portion of BASE24 R6.V10 in a separate location. This method does not interfere with the BASE portion of the existing BASE24.

    Reference 1: BASE24 6.0 Install Guide

    Reference 2: Section 6.1 Binding IMNI and IMNSC code under BASE24 R6.V10 of BASE24 IMNI/IMNSC Installation Guide V7.

2) BASE24 TSS release 9.2 SP6.
   Reference: HP Nonstop and HP NonStop Install Guide - BASE24-eps and TSS Release 1.0 Version 09.2

3) Convert to AKB mode.

Phase 1

1) Confirm that all the prerequisites for Phase 1 are satisfied;

2) Switch to the new BASE24 IMNi/IMNsc and TSS systems and start them up;

3) Perform key exchanges with single length KEK and ensure successes.

4) Make configuration changes according to Section 2.1-Phase 1 TSS configuration in the Addendum of ACI KEK Change Procedures for TDES.

5) Perform key exchanges with double length KEK and ensure successes.

6) Confirm the new system(s) are operating normally for online transactions.

Phase 2

1) Confirm that all the prerequisites for Phase 2 are satisfied.

2) Convert to AKB if required.

3) Make configuration changes according to Section 2.2 - Phase 2 TSS configuration in Addendum of ACI KEK Change Procedures for TDES;

4) Warmboot the new BASE24 system(s);

5) Perform double length working key exchanges and ensure successes;

6) Confirm the system(s) are operating normally for online transactions.

Rollback

When irresolvable issues/problems are encountered, the implementation needs to be rolled back to the previous system states.

Phase 1:

To restore the KEKs back to single length:

1) Set the KEYF records to point back to the previous key locator(s).
2) Warmboot the IMNi / IMNsc processes one process at a time.

Phase 2:

To restore the working keys back to single length:

1) Set the KEYF records to point back to the previous key locator(s).
2) Warmboot the IMNi / IMNsc processes one process at a time.

Note: The new TDES package of BASE24 works with both single length and double length working keys. It is at the customer's discretion whether to restore the system back to previous executables.

Report of Issues

1) For issues of severity 3 or lower, Open a SalesForce case using the following prefix on the subject of the case:

   INTERAC-TDES

2) Contact ACI Help24 during production implementation or for severity 1 or 2 issues.